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CODE:

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# Implementation of K-Nearest Neighbours (K-NN)

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Load the Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create K-NN model with K = 3
k = 3

knn = KNeighborsClassifier(n_neighbors=k)

# Train the model
knn.fit(X_train, y_train)

# Test the model
y_pred = knn.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("K-Nearest Neighbours (K-NN) Algorithm")
print("-----")
print("Value of K:", k)
print("Accuracy:", accuracy * 100, "%")

# Classify a new sample
new_sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = knn.predict(new_sample)

print("\nNew Sample:", new_sample[0])
print("Predicted Class:", iris.target_names[prediction[0]])
```

OUTPUT :

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K-Nearest Neighbours (K-NN) Algorithm
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Value of K: 3
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Accuracy: 100.0 %
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New Sample: [5.1, 3.5, 1.4, 0.2]
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Predicted Class: setosa
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